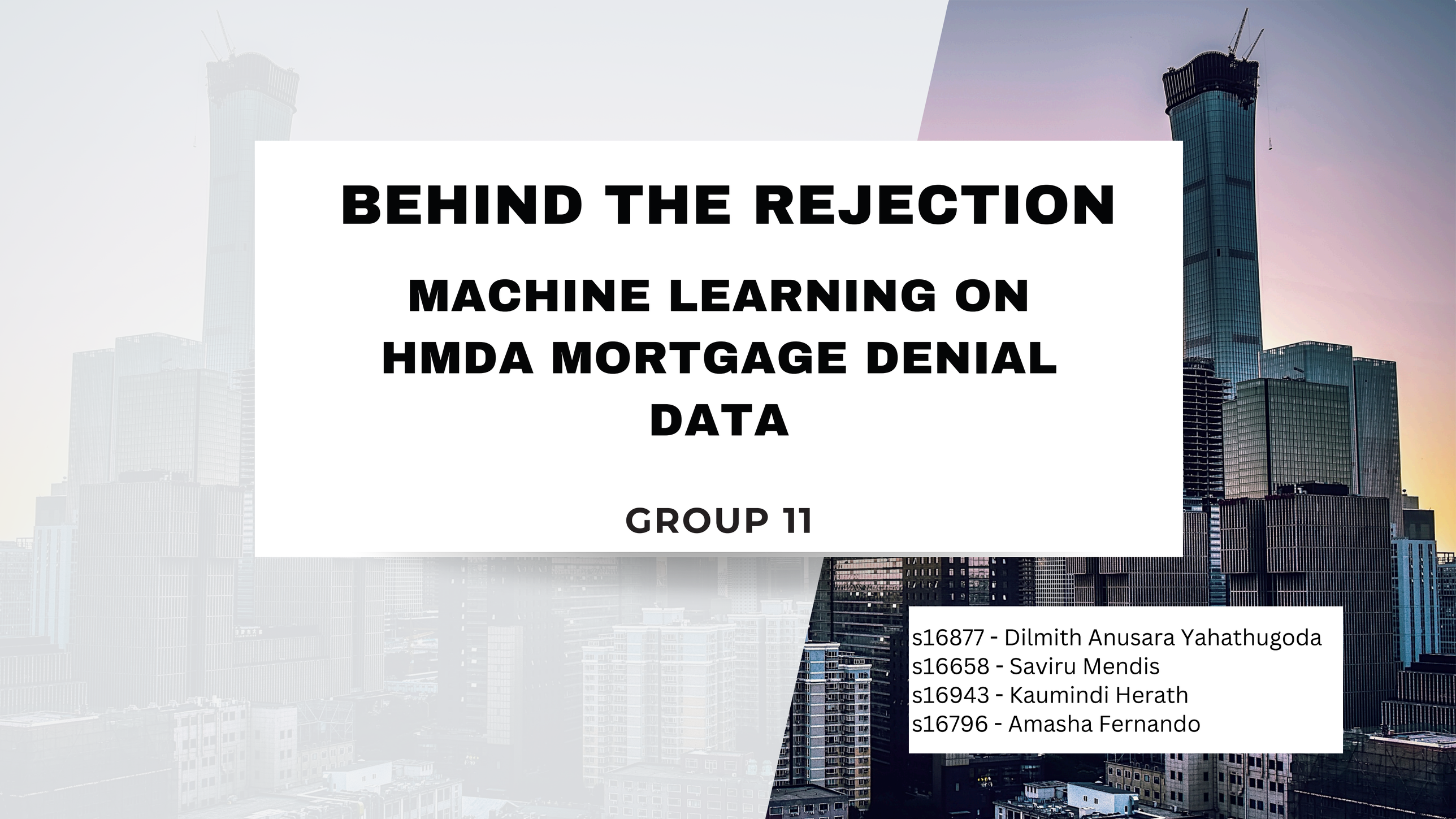
BEHIND THE REJECTION

MACHINE LEARNING ON HMDA MORTGAGE DENIAL DATA

GROUP 11



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PREDICTING MORTGAGE LOAN DENIALS

Using machine learning to understand why home loan applications get denied, and which factors matter most.

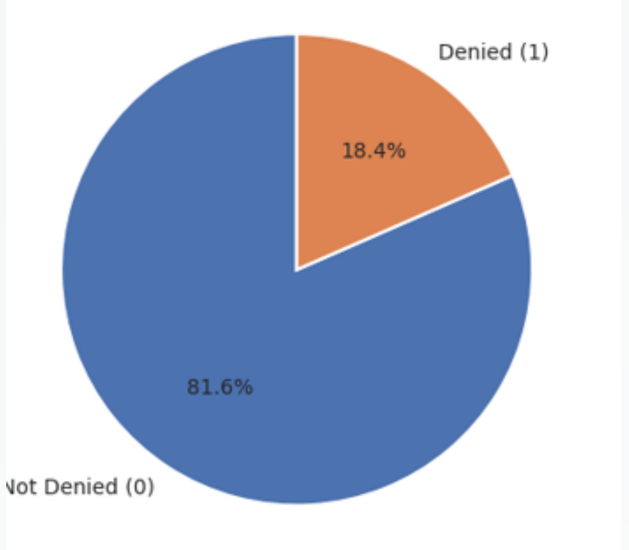
What We Are Doing

- Predict whether a mortgage application will be denied or approved
- Find out which factors drive that decision
- Use real 2024 Texas home-purchase loan data reported under HMDA (Home Mortgage Disclosure Act)

462,138 loan applications

99 → 42 features, after cleaning

18.4% of applications denied



Missing Value Handling (EDA Recap)

Types Identified

- **Not Reported** : lender did not provide the information (e.g., income, DTI)
- **Exempt** : lender was legally allowed not to report it. (e.g., property value, LTV)

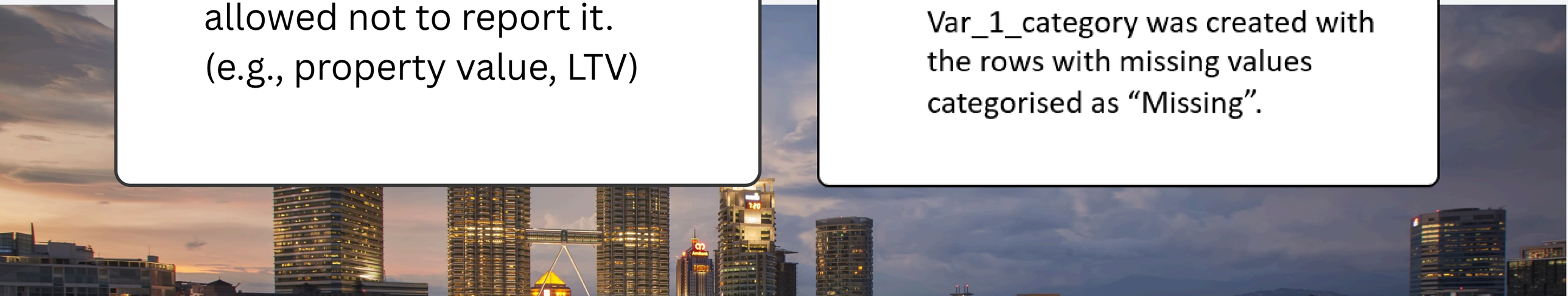
Finding

Missing values had different meanings depending on the variable.

EDA decision

Preserved the missingness information using categories.

- If Var_1 had missing values, Var_1_category was created with the rows with missing values categorised as “Missing”.



Outlier Detection (EDA Recap)

Approach

- Removed only records with impossible relationships.
- Ex: Property value: $1.05 billion
- Loan amount: $5,000
- Assume as data entry error

Finding

- Extreme values were not always errors.
- In mortgage data, expensive properties and large loans can be genuine.
- Therefore, we checked whether values were logically consistent rather than removing all extreme observations

Decision: removed inconsistent records only.

Why not cluster wise modeling ??

Splitting the dataset into 8 clusters reduces the no of observations for training and testing for each model.

Smaller clusters may produce unstable estimates and increase the risk of overfitting.

We lack the computational resources to do cluster wise modeling

Divide our Variables in to 4 groups

Applicant Characteristics

Who is applying ??

- **Variables:** derived_sex, derived_race, derived_ethnicity, applicant_age, co-applicant_age, debt_to_income_ratio, applicant_credit_score_type, co-applicant_credit_score_type

Loan Characteristics

What are they applying for?

- **Variables**: loan_type, loan_term, loan_term_category, lien_status, reverse_mortgage, interest_only_payment, balloon_payment, negative_amortization, open-end_line_of_credit, business_or_commercial_purpose, conforming_loan_limit.

Property Context Characteristics

Covers property details and location-related information

- **Variables:** derived_dwelling_category, occupancy_type, property_value_category,manufactured_home_secured_property_type, manufactured_home_land_property_interest, total_units, tract_population, tract_minority_population_percent, ffiec_msa_md_median_family_income,

Application Process Characteristics

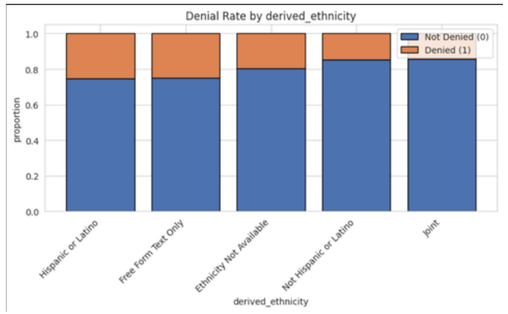
How the application is submitted and processed?

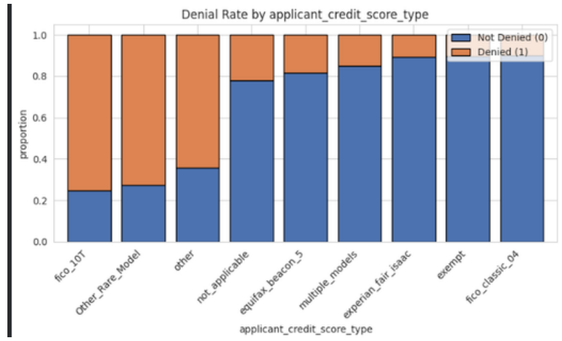
- **Variables**: submission_of_application, aus-1, income_was_missing, institution_exempt_flag.

Applicant and loan variables were all significantly related to approval

Applicant Characteristic

•Applicant Profile all the variables have significant relationship with loan approval.

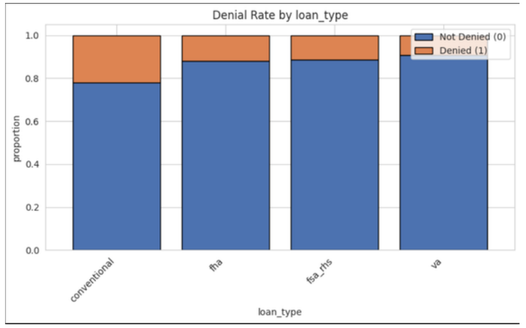


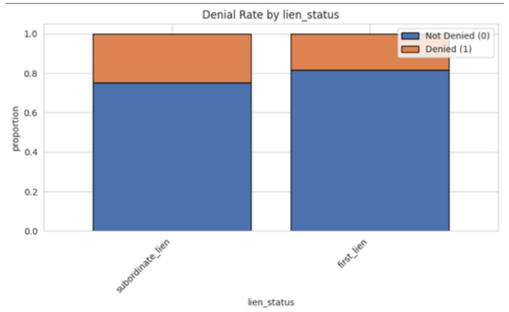


Variable	P-Value	Significant
debt_to_income_ratio	0	Yes
applicant_credit_score_type	0	Yes
derived_ethnicity	0	Yes
derived_race	0	Yes
derived_sex	0	Yes
applicant_age	6.4977E-294	Yes

Loan Characteristic

•Loan characteristics all the variables have significant relationship with loan approval.



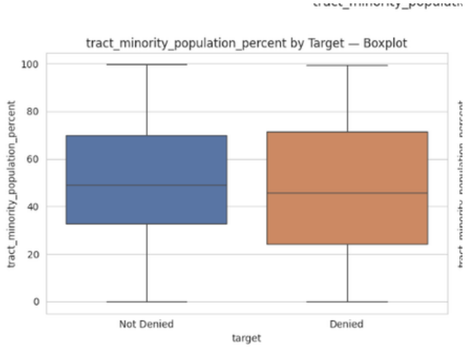


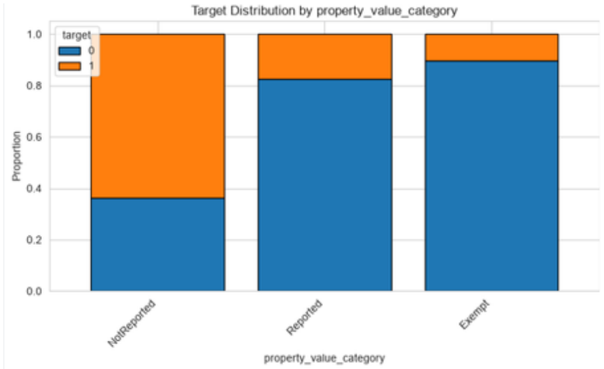
Variable	P-Value	Significant
loan_type	0	Yes
loan_term_category	6.2774E-266	Yes
business_or_commercial_purpose	4.0125E-151	Yes
interest_only_payment	2.5686E-105	Yes
balloon_payment	1.19566E-83	Yes
negative_amortization	4.52025E-79	Yes
conforming_loan_limit	5.22556E-60	Yes

Property and process variables were mostly significant too

Property Context Characteristics

all the variables have significant relationship with loan approval.



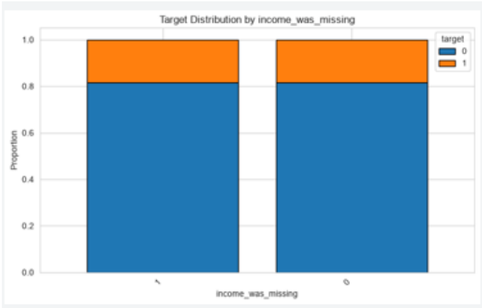


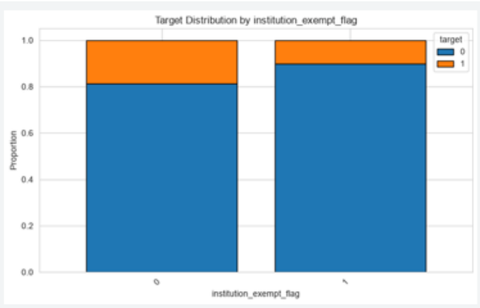
Variable	P-Value	Significant
tract_to_msa_income_percentage	0	Yes
tract_population_bin	0	Yes
tract_owner_occupied_units_bin	0	Yes
ffiec_msa_md_median_family_inco	0	Yes
tract_minority_population_percent	0	Yes
tract_one_to_four_family_homes_bi	0	Yes
tract_median_age_of_housing_units	0	Yes

Variable	P-Value	Significant
manufactured_home_secured_prop	0	Yes
manufactured_home_land_property	0	Yes
derived_dwelling_category	0	Yes
property_value_category	0	Yes
occupancy_type	0.000000000000	Yes
total_units	0.000000000000	Yes

Applicant Process

Most application process variables are statistically significant. Except income missing which is not significant.





Variable	P-Value	Significant
submission_of_application	0.0000000000000	Yes
institution_exempt_flag	0.0000000000000	Yes
income_was_missing	0.9798827	No

Data Preparation for Modeling

Missing Values

- Numerical missing values → Median Imputation
- Reason: Robust to skewed data and computationally efficient

Feature Processing

- Categorical variables → One-hot Encoding
- Numerical variables → Standardization

Class Imbalance

- Class distribution: Not Denied: 81.56% , Denied: 18.44%
- Applied Class Weighting

Why Class Weighting Instead of Other Methods

- SMOTE: Distance-based synthetic data may create unrealistic loan profiles with many categorical variables
- Undersampling: May remove useful information from the majority class.
- Class Weighting: Handles imbalance while keeping the original dataset unchanged.

Linear Models

Models Used:

- **Logistic Regression**
- **Ridge Logistic Regression**
- **Lasso Logistic Regression**
- **Elastic Net Logistic Regression**

Purpose:

- **Provides an interpretable baseline model.**

Logistic Regression

	Test	Train
Accuracy	0.8599	0.8604
Recall	0.7409	0.7380
F1-score	0.6602	0.6603
ROC-AUC	0.8817	0.8816

- ROC-AUC of 0.8816 indicates good classification ability.
- Recall of 73.8% , most denied applications were detected.
- Similar training/testing results mean good generalization.

Ridge Model

	Test	Train
Accuracy	0.8599	0.8605
Recall	0.7408	0.7379
F1-score	0.6602	0.6603
ROC-AUC	0.8817	0.8816

- Adds L2 penalty to shrink large coefficients keep all variables.
- Hyperparameter Tuning selected : C=10000
- Indicating that the model performed best with very weak regularization.
- L2 regularization provided no significant performance improvement.

Lasso Model

	Test	Train
Accuracy	0.8597	0.8602
Recall	0.7404	0.7383
F1-score	0.6597	0.6601
ROC-AUC	0.8815	0.8815

- Adds L1 penalty to shrink coefficients to zero .
- Hyperparameter Tuning selected: C=10000
- Large C shows weak regularization was preferred.
- L1 regularization provided no significant performance improvement

ElasticNet Model

	Test	Train
Accuracy	0.8597	0.8602
Recall	0.7402	0.7383
F1-score	0.6597	0.6600
ROC-AUC	0.8815	0.8815

- Combines L1 + L2 regularization.
- L1 feature selection
- L2 coefficient stability

- Hyperparameter Tuning selected : C=10000
- L1_ratio=0.2

- Large C shows weak regularization was preferred.

- Elastic Net regularization did not provide significant performance improvement.

Bagging Models

Models Used:

- **Random Forest**

Purpose:

- **Combines multiple decision trees to improve prediction stability.**
- **Captures complex relationships and feature interactions**

Random Forest

What is RF: Ensemble of decision trees (bagging) — bootstrapped samples + random feature subsets per tree, final prediction by majority vote. Handles non-linear relationships and feature interactions natively, but requires numeric input.

Encoding:

- Nominal (no order) → One-Hot Encoding (e.g. loan_type, derived_race, derived_dwelling_category)
- Ordinal (has order) → Rank/Integer Encoding (e.g. debt_to_income_ratio, total_units)

Missing values: sklearn's RF can't handle NaN directly → residual gaps median-imputed; missingness flags (e.g. property_value_category) kept as separate features, since missingness itself is informative in HMDA data.

Class imbalance: ~18% denial rate → class_weight='balanced'; PR-AUC used over accuracy/ROC-AUC as the primary metric.

Hyperparameters & Test Performance

Tuned via RandomizedSearchCV (single stratified validation split, scoring = average_precision):

Parameter	Value
n_estimators	300
max_depth	30
max_features	0.3
min_samples_leaf	9
min_samples_split	8

Test set results:

ROC-AUC: 0.9127
PR-AUC: 0.8308

Class	Precision	Recall	F1-score	Support
0 (Approved)	0.94	0.95	0.95	75,865
1 (Denied)	0.77	0.73	0.75	17,083
Accuracy			**0.91**	**92,948**
Macro avg	0.85	0.84	0.85	92,948
Weighted avg	0.91	0.91	0.91	92,948

Feature Importance (Permutation Importance)

Note:

permutation importance used instead of built-in impurity importance, which is biased toward high-cardinality/continuous features.

Feature	Importance
debt_to_income_ratio	0.3394
aus-1_not_applicable	0.1056
loan_type_conventional	0.0224
loan_to_value_ratio	0.0219
income	0.0188
property_value	0.0133
loan_term	0.0097
derived_dwelling_category_single_family_manufactured	0.0092
loan_amount	0.0064
loan_type_va	0.0057
property_value_category_NotReported	0.0051
derived_dwelling_category_single_family_site_built	0.004
applicant_credit_score_type_equifax_beacon_5	0.0037
manufactured_home_land_property_interest_not_applicable	0.0032
applicant_credit_score_type_other	0.0031
tract_owner_occupied_units	0.0031
loan_to_value_ratio_category_Reported	0.0029
loan_type_fha	0.0028
loan_to_value_ratio_category_NotReported	0.0027
ffiec_msa_md_median_family_income	0.0027

Fairness Analysis — Race

Disparate Impact Ratio (four-fifths rule — groups below 0.80 FAIL):

Group	Disparate Impact	n	TPR (Recall)	FPR	PPV (Precision)
Native Hawaiian or Other Pacific Islander	0.607	239	0.851	0.166	0.769
Free Form Text Only	0.794	230	0.75	0	1
American Indian or Alaska Native	0.798	800	0.774	0.071	0.798
Race Not Available	0.829	16,861	0.779	0.069	0.766
2 or more minority races	0.846	296	0.693	0.05	0.825
Black or African American	0.85	8,091	0.695	0.05	0.819
White	0.891	54,477	0.737	0.051	0.755
Joint	0.936	2,611	0.69	0.038	0.748
Asian	1	9,370	0.58	0.013	0.835

Max TPR gap: 0.271 Max FPR gap: 0.166

Fairness Analysis — Ethnicity & Sex

Ethnicity — Disparate Impact:

Group	Disparate Impact	n	TPR	FPR	PPV
Hispanic or Latino	0.867	26,617	0.782	0.07	0.787
Ethnicity Not Available	0.933	13,382	0.74	0.058	0.755
Free Form Text Only	0.95	34	0.667	0	1
Not Hispanic or Latino	0.999	48,378	0.687	0.038	0.757
Joint	1	4,537	0.706	0.04	0.744

Max TPR gap: 0.115 Max FPR gap: 0.070

Sex — Disparate Impact:

Group	Disparate Impact	n	TPR	FPR	PPV
Female	0.89	19,039	0.784	0.059	0.811
Male	0.954	32,486	0.722	0.053	0.758
Sex Not Available	0.966	6,432	0.728	0.047	0.774
Joint	1	34,991	0.701	0.043	0.741

Max TPR gap: 0.083 Max FPR gap: 0.016

Boosting Models

Models Used:

- **Xgboost**
- **CatBoost**
- **LightGBM**

Purpose:

- **Designed for high-performance prediction on large tabular datasets.**
- **Handles complex feature interactions efficiently.**

XGBoost

XGBoost fits this data natively -no scaling, handles missing values itself

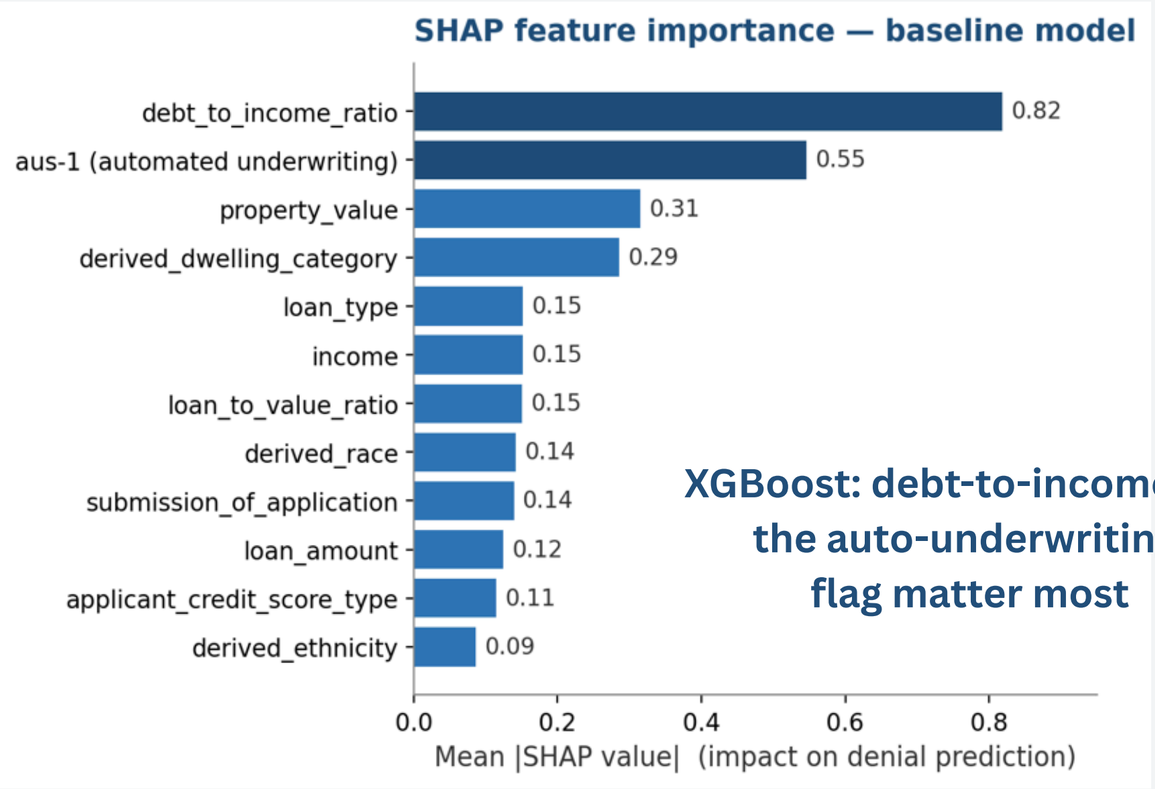
How did we handle class imbalance?

Instead of changing the data by oversampling or undersampling, we used XGBoost's built-in scale_pos_weight option (set to 4.44), which gives the minority (denied) class more weight during training.

Tuned model performance

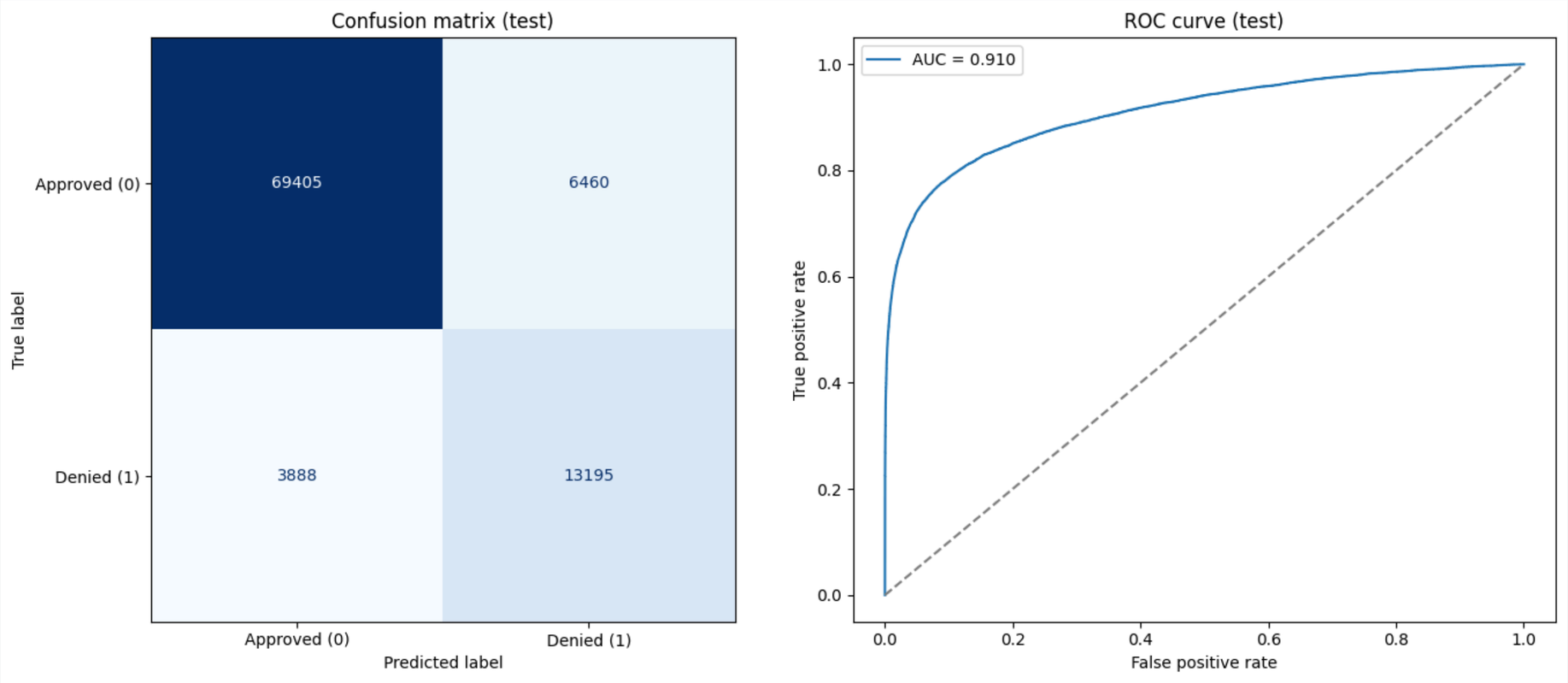
```
Accuracy : 0.8887
Precision: 0.6713
Recall   : 0.7724
F1-score : 0.7183
ROC-AUC  : 0.9103
PR-AUC   : 0.8216
```

	precision	recall	f1-score	support
0	0.95	0.91	0.93	75865
1	0.67	0.77	0.72	17083
accuracy			0.89	92948
macro avg	0.81	0.84	0.82	92948
weighted avg	0.90	0.89	0.89	92948



XGBoost: debt-to-income and the auto-underwriting flag matter most

How Well Does XGBoost Predict Denials?



- **The confusion matrix shows that on the 92,948 test applications the XGBoost model correctly identified 13,209 denials and 69,557 approvals, while missing 3,874 real denials and wrongly flagging 6,308 approved loans as denied meaning it catches about 77% of true denials.**
- **The ROC curve rises steeply toward the top-left corner with an AUC of 0.9111, showing the model separates denied from approved applications very well across all decision thresholds.**

Fairness Analysis

Evaluate whether the model makes equitable predictions across different demographic groups

> By race (measured with Disparate Impact; ≥ 0.80 passes the four-fifths rule)

- Five groups fall below 0.80 and are treated unfairly: Native Hawaiian/Pacific Islander 0.59, Free Form Text 0.67, American Indian/Alaska Native 0.75, 2+ minority races 0.77, Black or African American 0.79 (Asian is the reference at 1.00).
- Biggest gap in catching denials between groups (recall/TPR gap): 0.375.

> By ethnicity

- All groups pass , the lowest is Hispanic or Latino at 0.84; recall gap only 0.15.

> By sex

- All groups pass, the lowest is Female at 0.87; recall gap only 0.08

- XGBoost gives good predictive performance, but the fairness check shows clear disparities by race, while ethnicity and sex are fair.
- Both predictive performance and fairness should be considered before using the model for real loan decisions.
- Random Forest was slightly fairer by race (minimum disparate impact 0.61 and a smaller recall gap of 0.27) than XGBoost (0.59 and 0.38), but both models fail the four-fifths rule for the same racial groups.

CatBoost

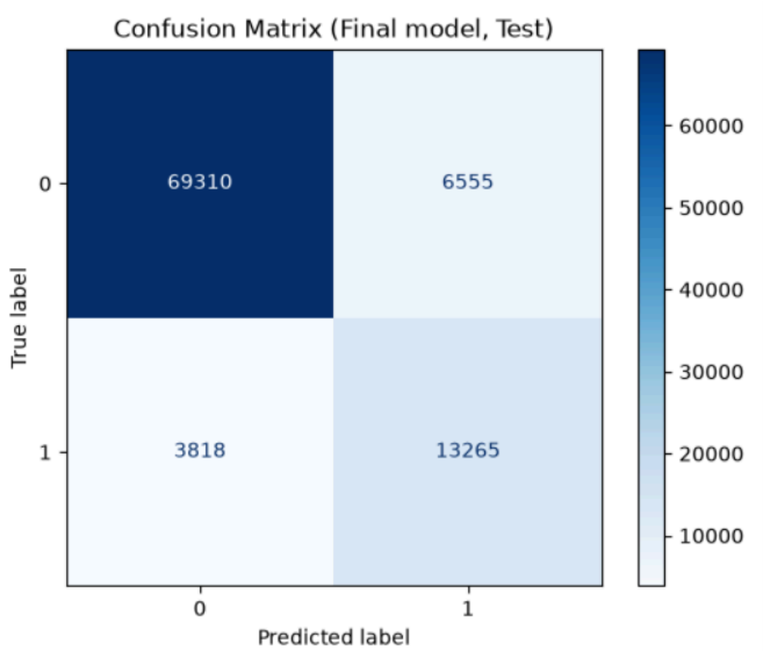
CatBoost (Categorical Boosting) is a machine learning algorithm based on gradient boosting that combines many small decision trees to make highly accurate predictions.

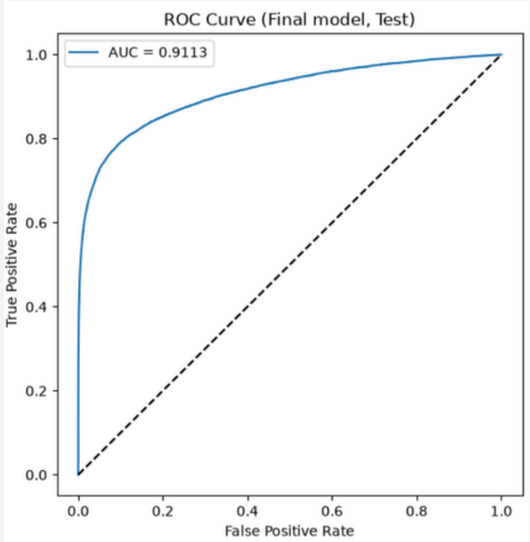
How did we handle class imbalance?

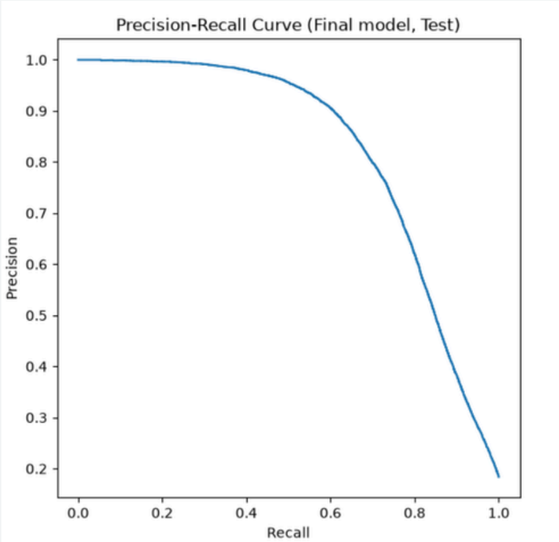
Instead of changing the data by oversampling or undersampling, we used CatBoost's built-in auto_class_weights='Balanced' option.

Tuned model performance

```
Accuracy : 0.8884
Precision: 0.6693
Recall   : 0.7765
F1 Score : 0.7189
ROC AUC  : 0.9113
PR AUC   : 0.8237

              precision    recall  f1-score   support

           0       0.95      0.91      0.93     75865
           1       0.67      0.78      0.72     17083

    accuracy                           0.89     92948
   macro avg       0.81      0.85      0.82     92948
weighted avg       0.90      0.89      0.89     92948
```







SHAP importance

feature	importance
debt_to_income_ratio	0.64868
property_value	0.274563
aus-1	0.27267
loan_to_value_ratio	0.168789
loan_type	0.168733
income	0.164593
submission_of_application	0.154146
manufactured_home_secured_p	0.147648
derived_race	0.143163
derived_dwelling_category	0.119013
applicant_credit_score_type	0.109461

Fairness Analysis

Objective

Evaluate whether the model makes equitable predictions across different demographic groups.

Protected Group	Minimum DI	Four-Fifths Rule	Recall/FPR Gap	Interpretation
Race	**0.574**	❌ Failed	High (TPR = 24.3%, FPR = 22.1%)	Potential racial disparities detected.
Ethnicity	**0.84**	✓ Passed	Moderate	Fairness generally acceptable across ethnic groups.
Sex	**0.869**	✓ Passed	Small	Predictions were relatively equitable across sex groups.

Conclusion

- The CatBoost model achieved good predictive performance.
- Fairness analysis identified potential disparities across some racial groups.
- Both predictive performance and fairness should be considered before deploying the model in real-world loan approval systems.

LightGBM

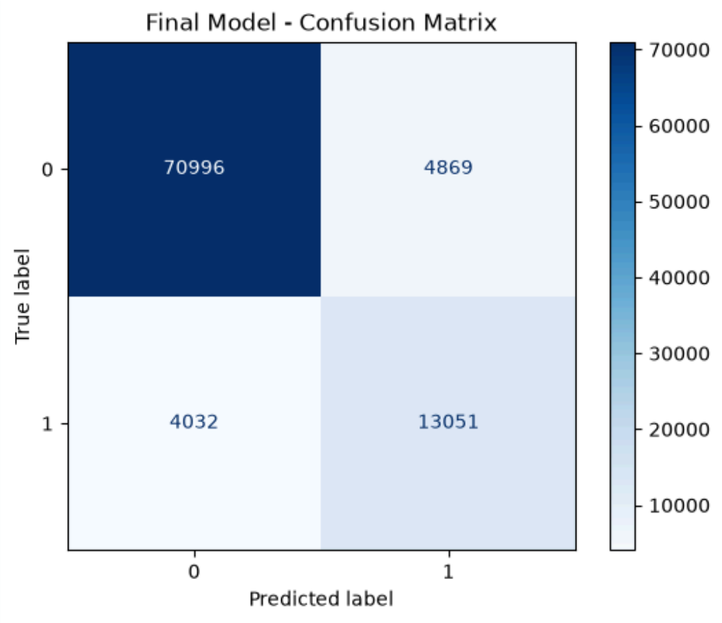
LightGBM (Light Gradient Boosting Machine) is a gradient boosting algorithm that builds decision trees efficiently using a leaf-wise growth strategy, enabling fast training and high predictive performance on large datasets.

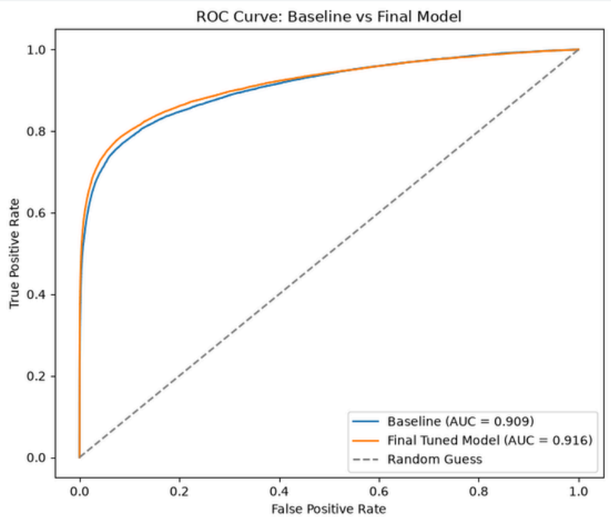
How did we handle class imbalance?

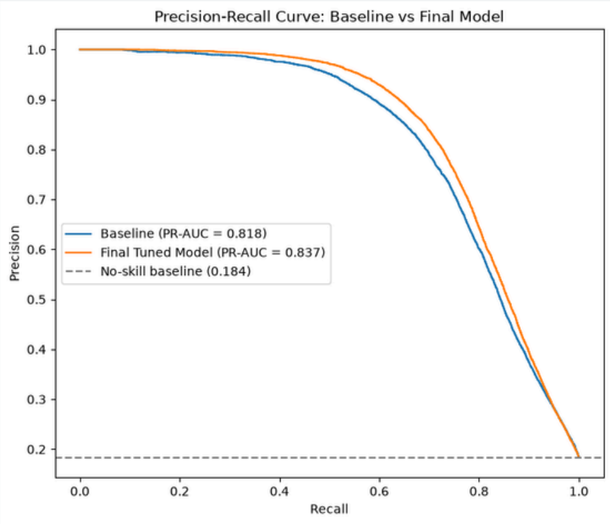
Instead of oversampling or undersampling, we used **class_weight = "balanced".** This automatically assigns higher weights to the minority class during training.

Tuned Model Performance

```
============================================================
FINAL MODEL (tuned + balanced)
============================================================
Accuracy : 0.9042
Precision: 0.7283
Recall   : 0.7640
F1 Score : 0.7457
ROC-AUC  : 0.9156
PR-AUC   : 0.8368

Classification Report
              precision    recall  f1-score   support

           0       0.95      0.94      0.94     75865
           1       0.73      0.76      0.75     17083

    accuracy                           0.90     92948
   macro avg       0.84      0.85      0.84     92948
weighted avg       0.91      0.90      0.91     92948
```







Feature importance values(top 10)

Feature	Importance
income	16745
property_value	12889
loan_to_value_ratio	12366
tract_minority_population_percent	11111
tract_population	10686
tract_to_msa_income_percentage	10491
tract_owner_occupied_units	10186
tract_one_to_four_family_homes	10133
loan_amount	9932
tract_median_age_of_housing_units	9146

Fairness Analysis

Objective

Evaluate whether the LightGBM model makes equitable predictions across different demographic groups.

Protected Group	Minimum DI	Four-Fifths Rule	Max Recall Gap	Max FPR Gap	Interpretation
Race	**0.612**	❌ Failed	**24.40%**	**14.10%**	Several racial groups received substantially lower approval rates than the most-favored group, indicating potential racial disparity.
Ethnicity	**0.85**	✓ Passed	**14.60%**	**4.50%**	Approval rates were generally balanced, with no evidence of significant disparate impact.
Sex	**0.875**	✓ Passed	**8.60%**	**2.80%**	Approval rates and prediction performance were relatively consistent across sex groups.

Conclusion

- The LightGBM model achieved strong predictive performance (Accuracy = 90.42%, ROC-AUC = 0.9156).
- Fairness analysis revealed the largest disparities across racial groups, while ethnicity and sex showed comparatively smaller differences.
- Both predictive performance and fairness should be evaluated before deploying the model in real-world loan approval systems.

Summary

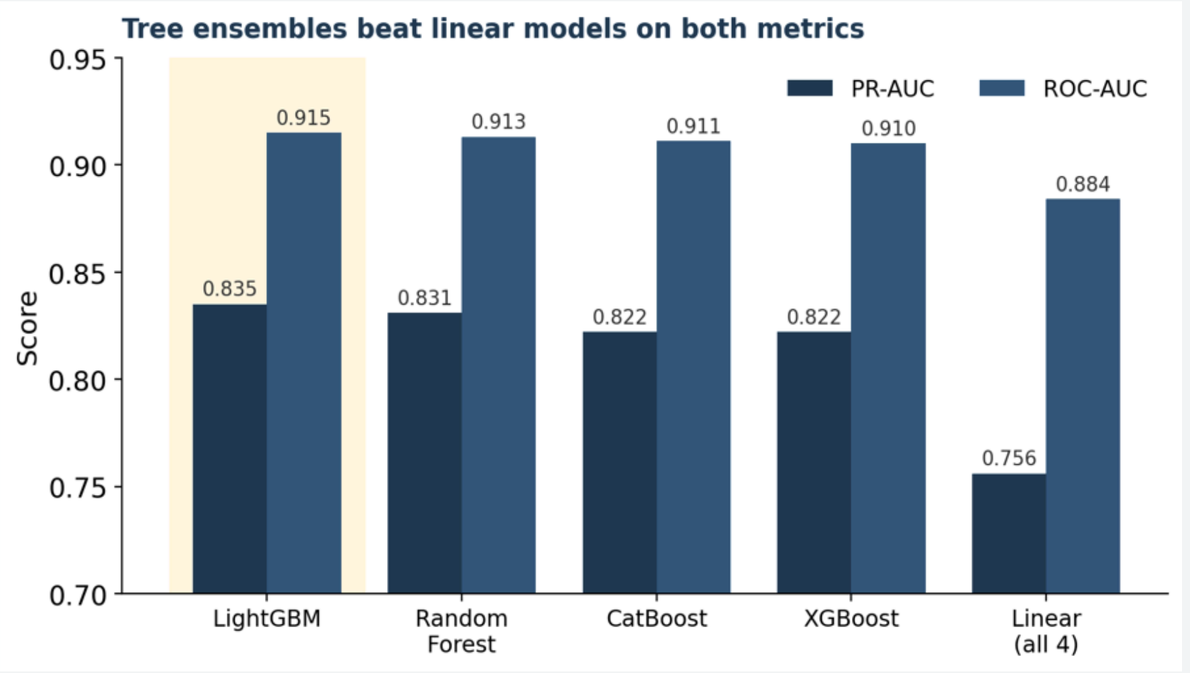
Model	ROC-AUC	PR-AUC	F1-score	Recall	Race	Ethnicity	Sex
Logistic Regression	0.8816	0.7489	0.6603	0.738	✗	✗	✓
Random Forest	0.9127	0.8308	0.75	0.73	✗	✓	✓
XGBoost	0.9111	0.8237	0.7218	0.7732	✗	✓	✓
CatBoost	0.9113	0.8237	0.7189	0.7765	✗	✓	✓
LightGBM	**0.9156**	**0.8368**	**0.7457**	**0.764**	✗	✓	✓

MODEL COMPARISON

Predictive Performance : Trees Outperform Linear Models

Tree ensembles beat linear models by a consistent, non-trivial margin on both metrics.

Model	PR-AUC	ROC-AUC
LightGBM	**0.835**	**0.915**
Random Forest	0.831	0.913
CatBoost	0.822	0.911
XGBoost	0.822	0.91
Logistic / Ridge / Lasso / ElasticNet	0.756	0.884

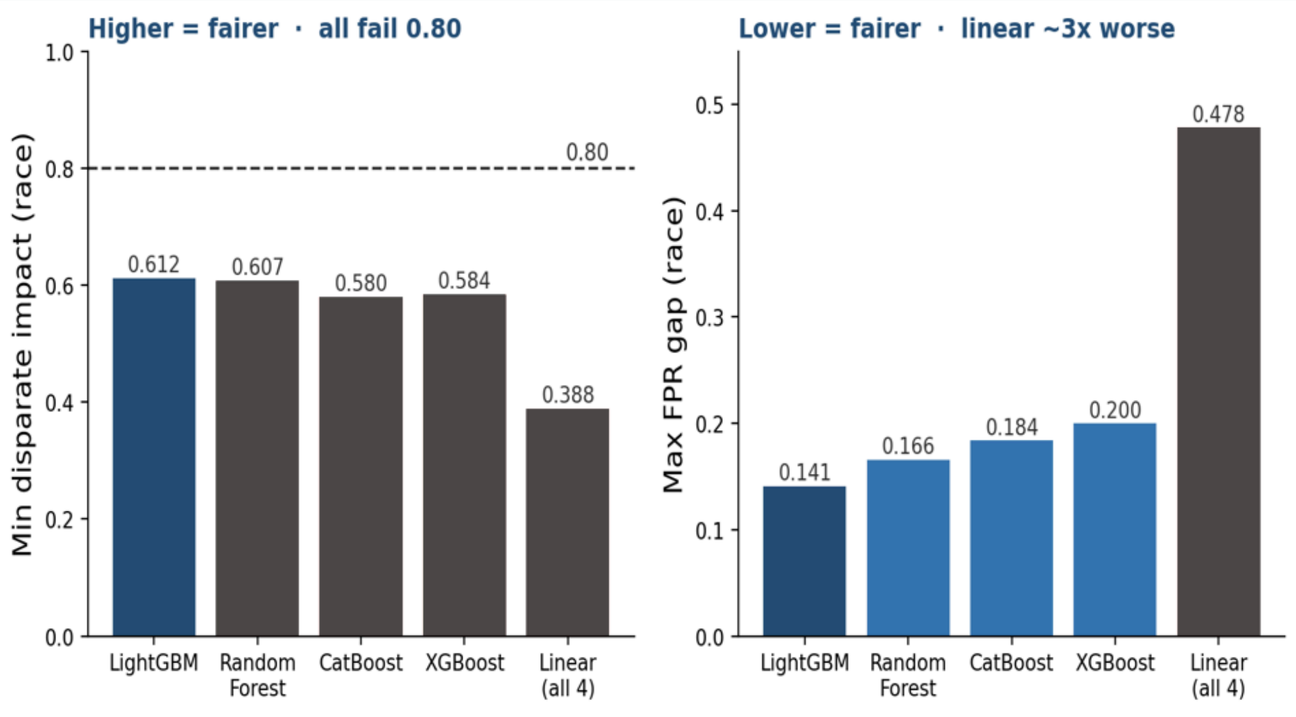


All four linear variants land within 0.0001 of each other, regularization type isn't the bottleneck, the linear decision boundary is.

Fairness - The Gap Widens, Not Narrows

Linear models aren't just less accurate — they're substantially less fair on the same split.

Model	Min DI (Race)	Max FPR Gap
LightGBM	**0.612**	**0.141**
Random Forest	0.607	0.166
CatBoost	0.58	0.184
XGBoost	0.584	0.2
Linear models (all 4)	~0.388	~0.478

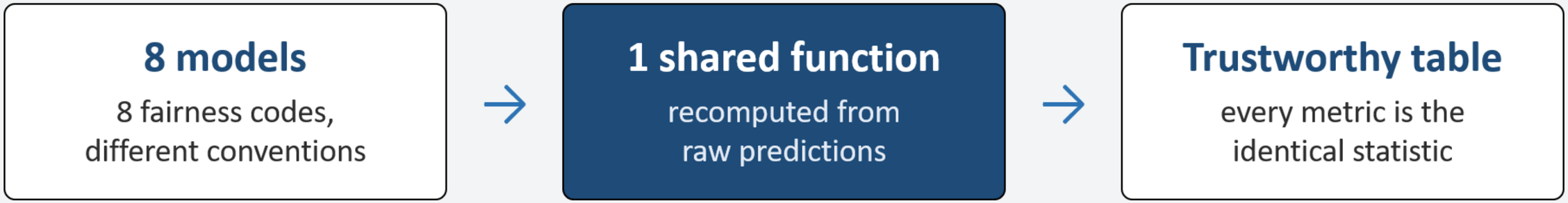


Linear disparate impact is ~1.5x worse than trees; FPR gap is ~3x worse.

Ensuring a Fair Comparison : One Definition, Applied Consistently

Each model had its own fairness code, written independently, a consolidated comparison required checking they all measured the same thing before trusting the table.

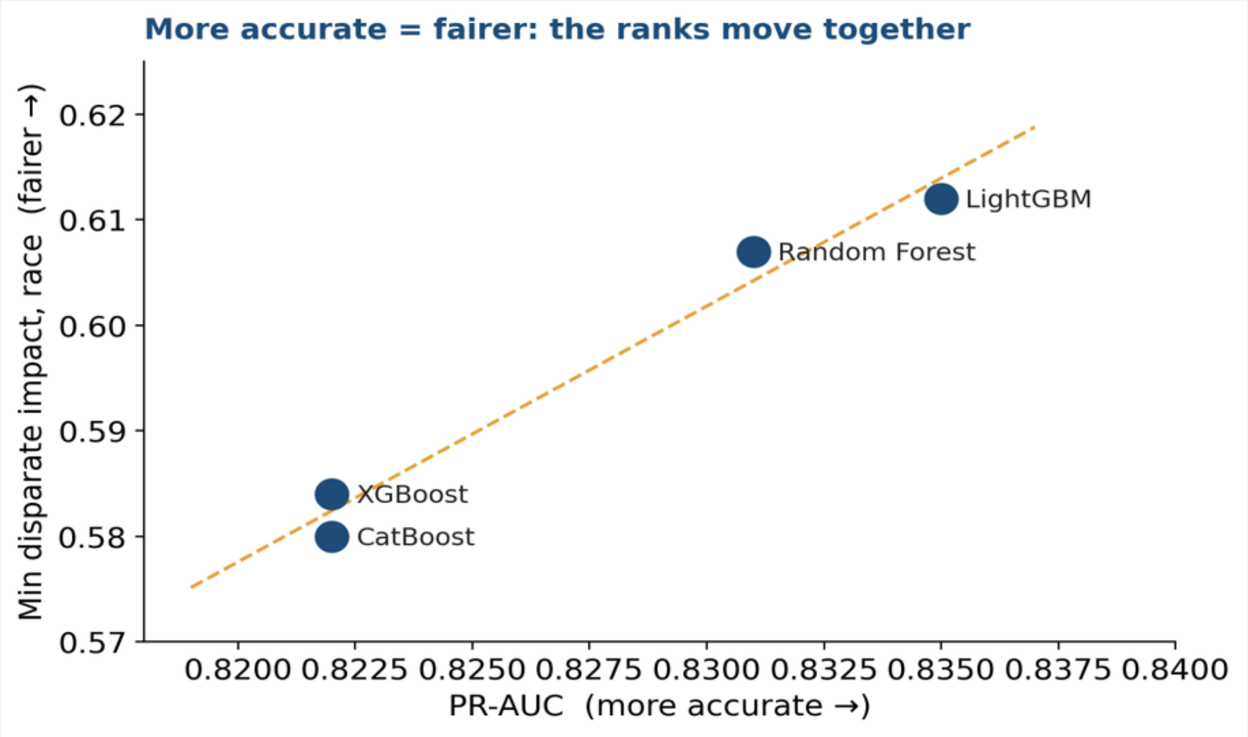
- One teammate's fairness function computed disparate impact from the **denial rate, not the approval rate ,** the opposite convention from the rest.
- We recomputed all 8 models' fairness metrics with **one shared function, from raw predictions,** rather than combining each model's own summary output.
- This ensures every number in this comparison is measuring the identical statistic.



No Accuracy - Fairness Trade-off Among Tree Models

Normally "most accurate" and "most fair" pull in different directions. Here, they don't.

Model	PR-AUC Rank	Fairness Rank (Race)
LightGBM	**1st**	**1st**
Random Forest	2nd	2nd
XGBoost	3rd	3rd
CatBoost	4th	4th



The real trade-off in this project is tree vs. linear : not accuracy vs. fairness within the trees.

Verdict

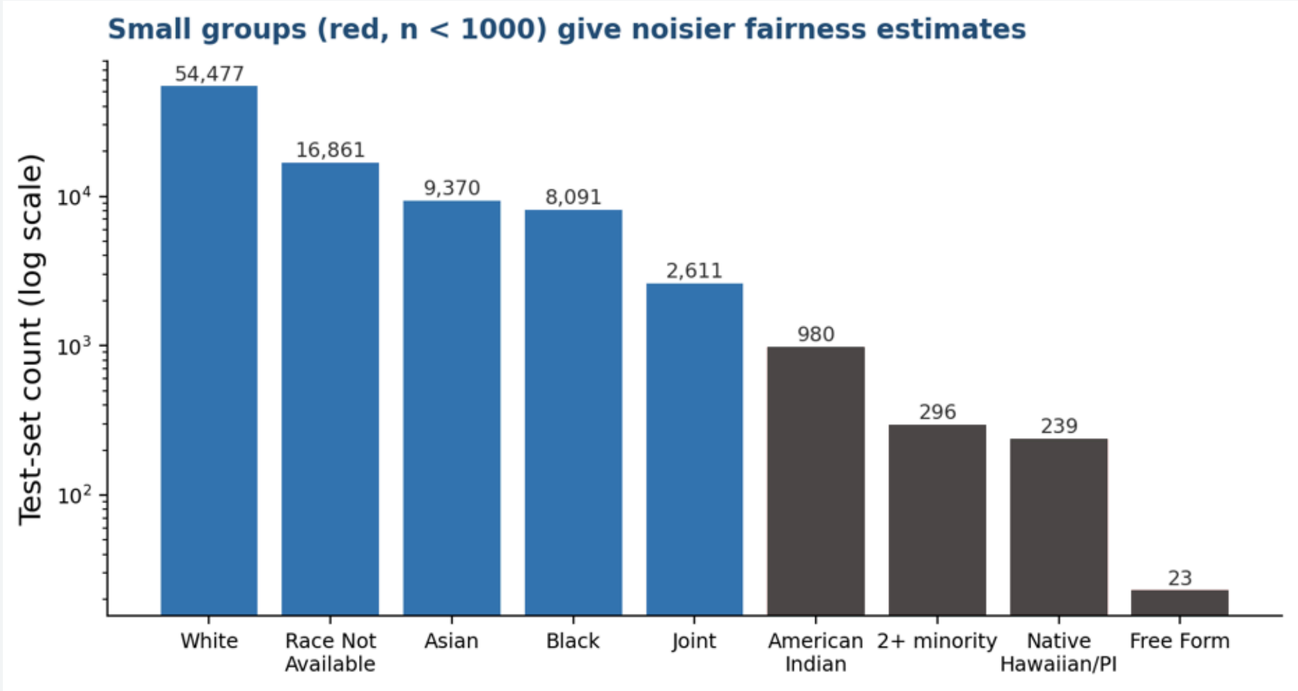
Recommend **LightGBM** : best on both axes, so there's no trade-off to defend.

★ **Best overall**	**LightGBM**	PR-AUC 0.835 and disparate impact 0.612 : best of all 8 models on BOTH axes.
Best on ethnicity / sex	**Random Forest**	Disparate impact 0.867 (ethnicity) and 0.890 (sex) : the fairest on those two attributes.
Avoid	**Any linear model**	Worse on every metric, with no meaningful difference between the four variants.

Limitations

- **No model passes the four-fifths rule (0.80) on race** : even LightGBM's 0.612 fails it

- **derived_race/derived_ethnicity/derived_sex were included as direct model inputs, not excluded** : results reflect direct + indirect effects combined



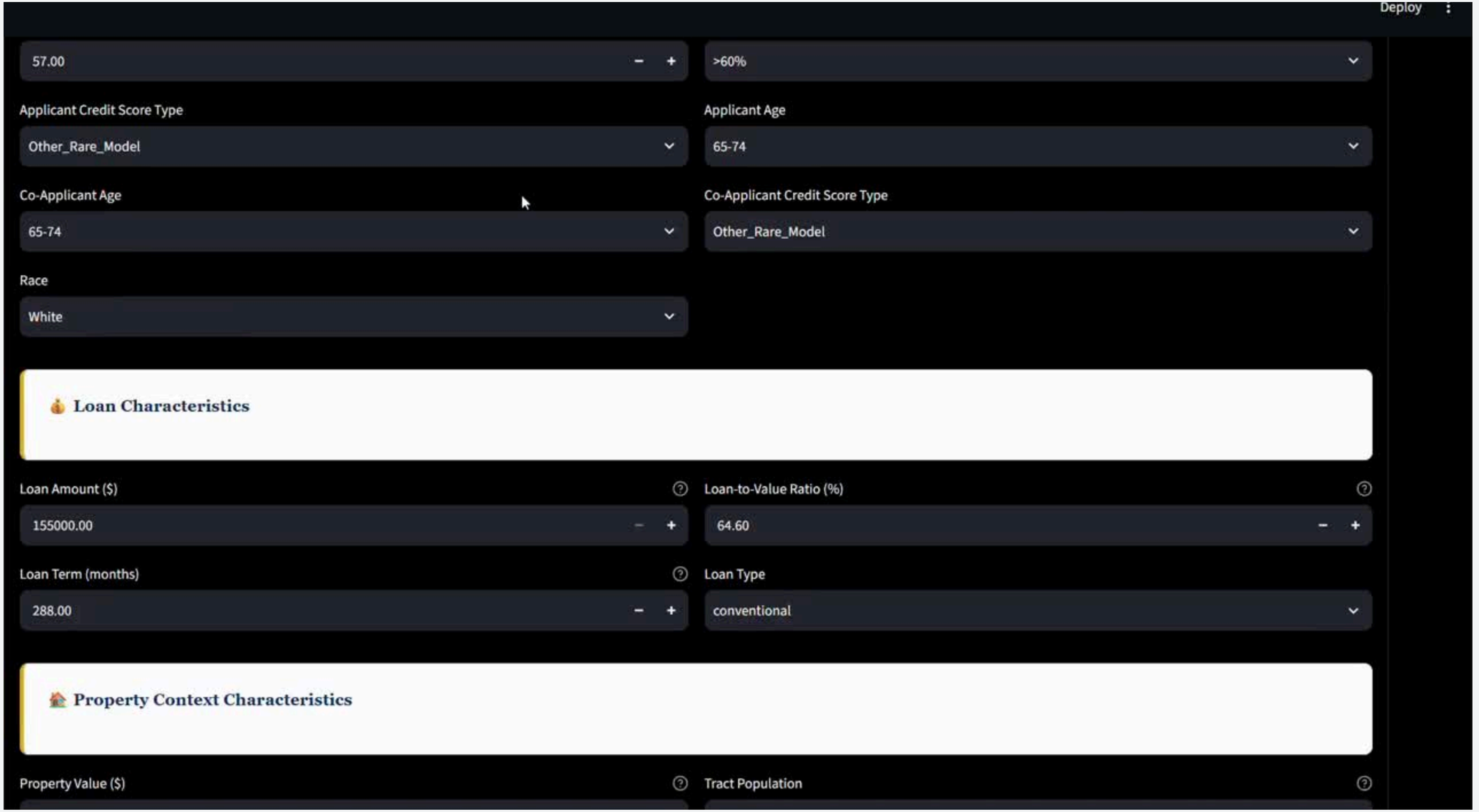
- **Small-n groups (e.g. n=239) carry noisier estimates than large groups (n=50,000+)** : treat low-n disparities as directional, not precise

Problems Encountered and How We Solved Them

Problem		Solution
Each model had its own fairness code, with different conventions	→	Recomputed all 8 with one shared function, from raw predictions
Denials are rare (18.4%) : models could ignore them	→	Used class weighting / scale_pos_weight (data kept unchanged)
No model is fair by race (all fail the 0.80 rule)	→	Measured it openly with disparate impact + TPR/FPR gaps
Protected attributes were used as direct inputs	→	Flagged that results mix direct + indirect effects
Small groups give noisy fairness numbers	→	Reported low-n disparities as directional, not precise
Linear models underperformed on every metric	→	Chose tree ensembles; kept linear only as a baseline

Suggestions to Overcome the Problems

- **Make the model fairer**: give small groups more weight, or use fairness rules in training. Push the race score above 0.80.
- **Test without race, ethnicity, and sex**: remove these fields and re-train. See how much unfairness they cause directly.
- **Get more data for small groups**: some groups are tiny (n=239). More data makes their numbers reliable.
- **Set the cut-off per group**: tune the decision line for each group. Catch real denials without wrong rejections.
- **Test on other states and years**: check the model works beyond 2024 Texas.
- **Keep a human in the loop**: let a person review close calls before any real decision.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>60%
Applicant Credit Score Type
Other_Rare_Model
Applicant Age
65-74
Co-Applicant Age
65-74
Co-Applicant Credit Score Type
Other_Rare_Model
Race
White
Loan Characteristics
Loan Amount ($)
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Loan-to-Value Ratio (%)
64.60
Loan Term (months)
288.00
Loan Type
conventional
Property Context Characteristics
Property Value ($)
Tract Population

THANK YOU